

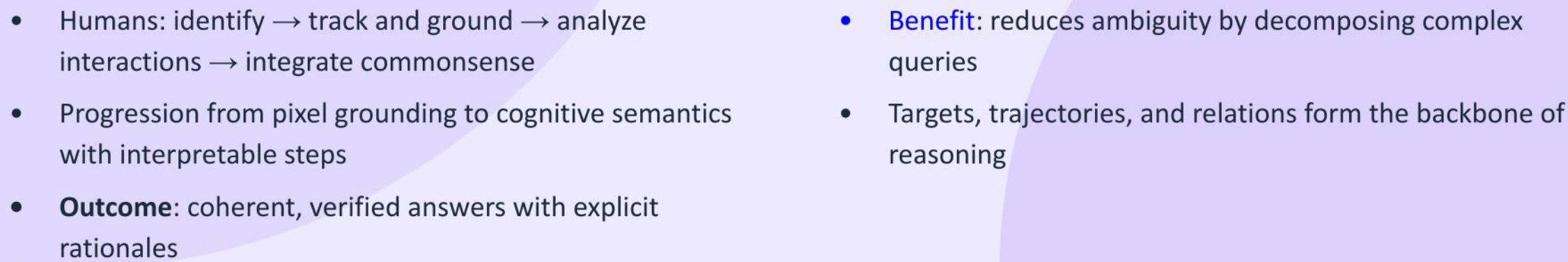


Video-of-Thought: Step-by-Step Video Reasoning from Perception to Cognition

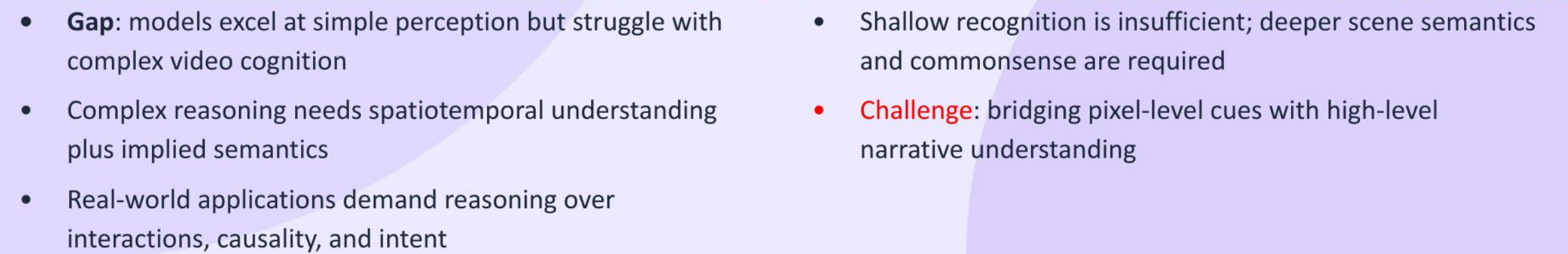
Hao Fei; Shengqiong Wu; Wei Ji; Hanwang Zhang; Meishan Zhang; Mong Li Lee; Wynne Hsu

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02 Human-like Multi-step Reasoning: Core Idea

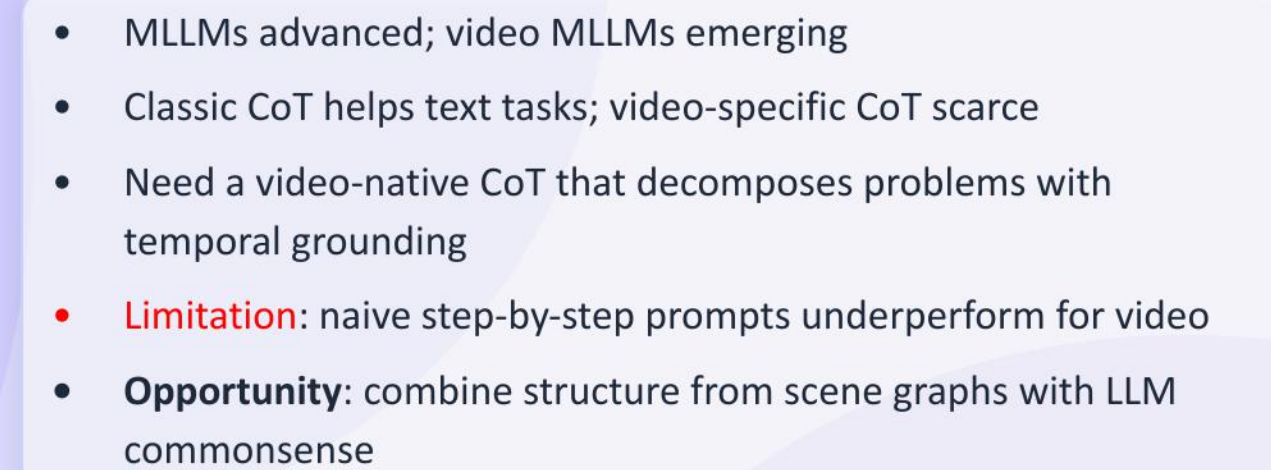


01 Motivation: From Perception to Complex Video Reasoning

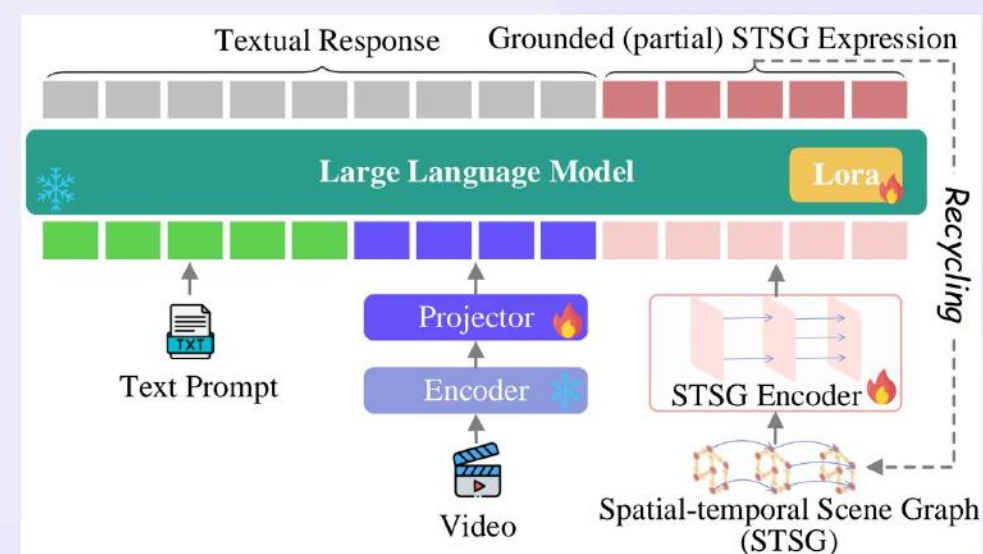


Lei et al., 2018; Bertasius et al., 2021; Zolfaghari et al., 2018; Lin et al., 2019; Caballero et al., 2017

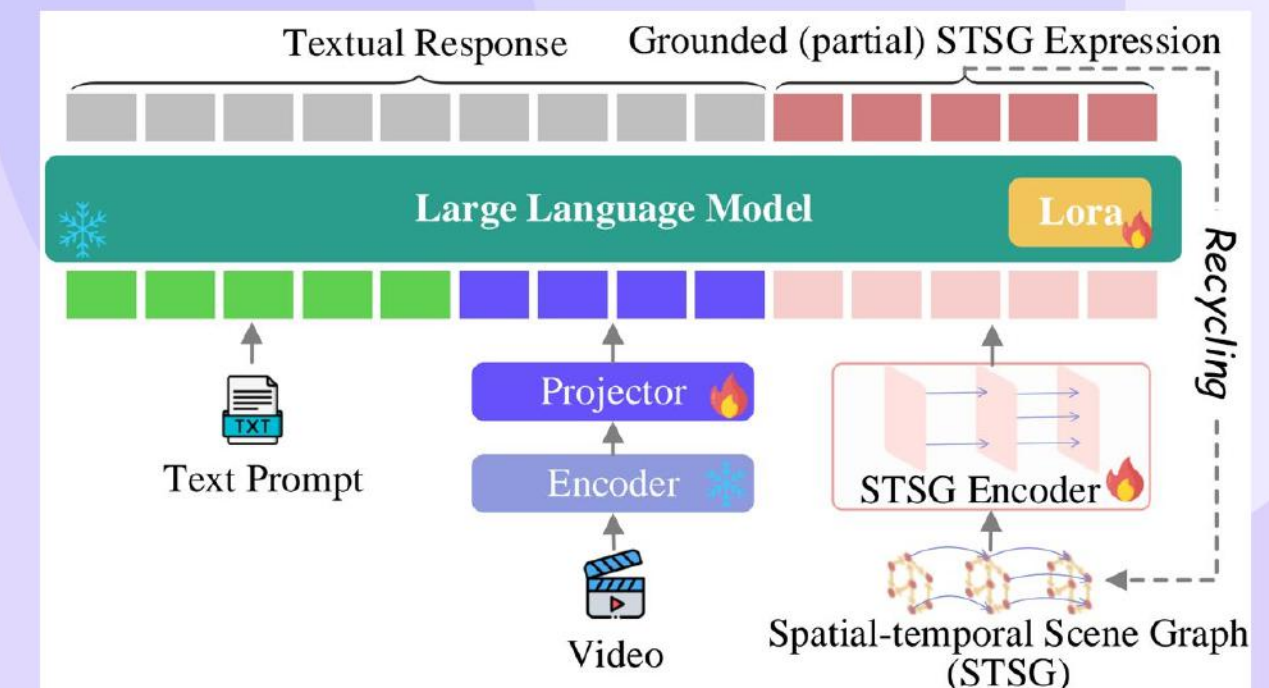
03 CoT for Video MLLMs is Under-explored



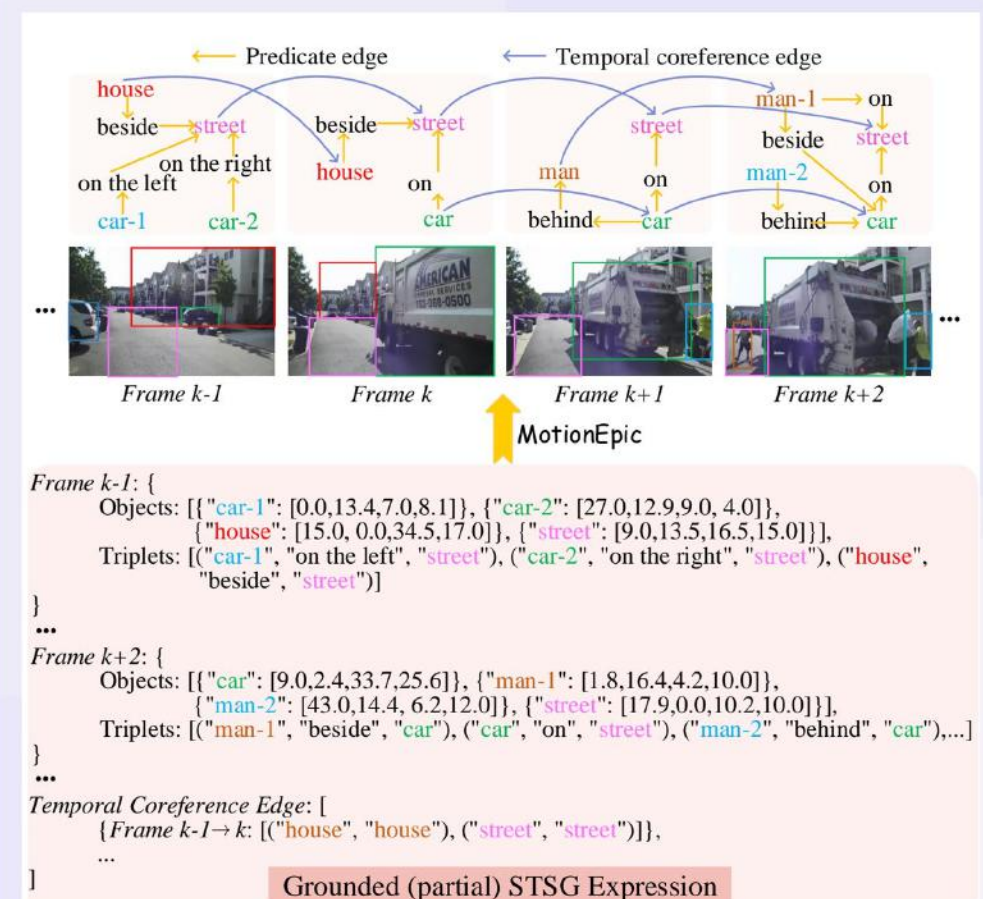
Zhang et al., 2023a; Maaz et al., 2023; Lin et al., 2023; Wei et al., 2022; Wang et al., 2022a



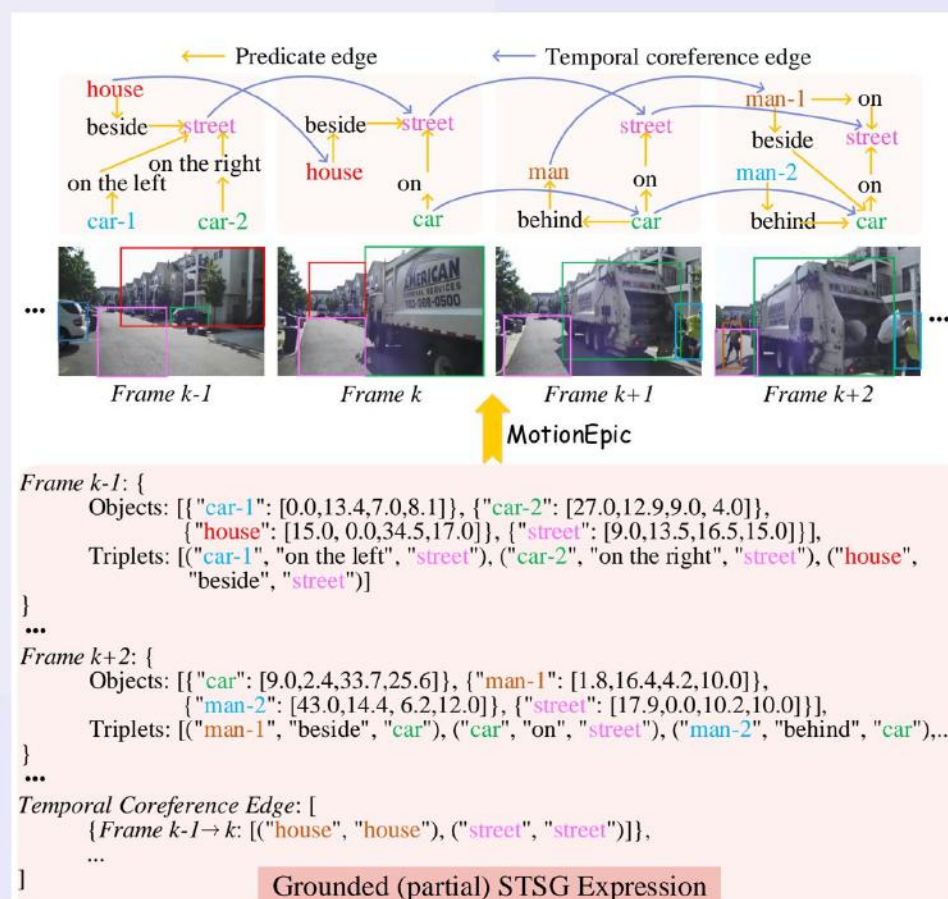
- **MotionEpic**: video MLLM that understands and generates STSGs
- **VoT**: chains video reasoning from grounding to cognition
- Grounding-aware tuning enables STSG-free downstream inference
- **Gain**: interpretable rationales through generated structure
- Unified pipeline for perception, interaction analysis, and answer verification



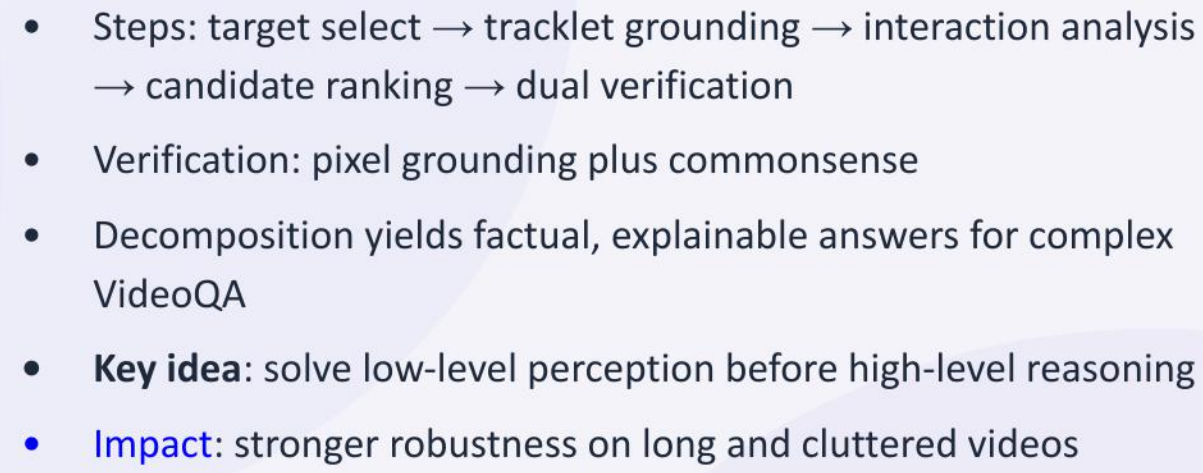
- Inputs: text, video, and STSG
- Backbone: Vicuna-7B; Video: ViT-L/14 plus Q-Former; STSG: recurrent Graph Transformer
- Aligned multi-source encodings for grounding-aware reasoning
- **Design**: structured nodes and relations augment dense visual tokens
- **Effect**: improved alignment between content and semantics



- STSG: per-frame scene graphs plus temporal coreference edges
- Nodes carry category, features, and bounding box; edges capture predicates and tracking
- Sampling and recurrent propagation improve efficiency and temporal continuity
- **Goal**: robust trajectories and relations for long-range reasoning
- **Risk**: over-sampling adds redundancy and noise; balanced rate preferred



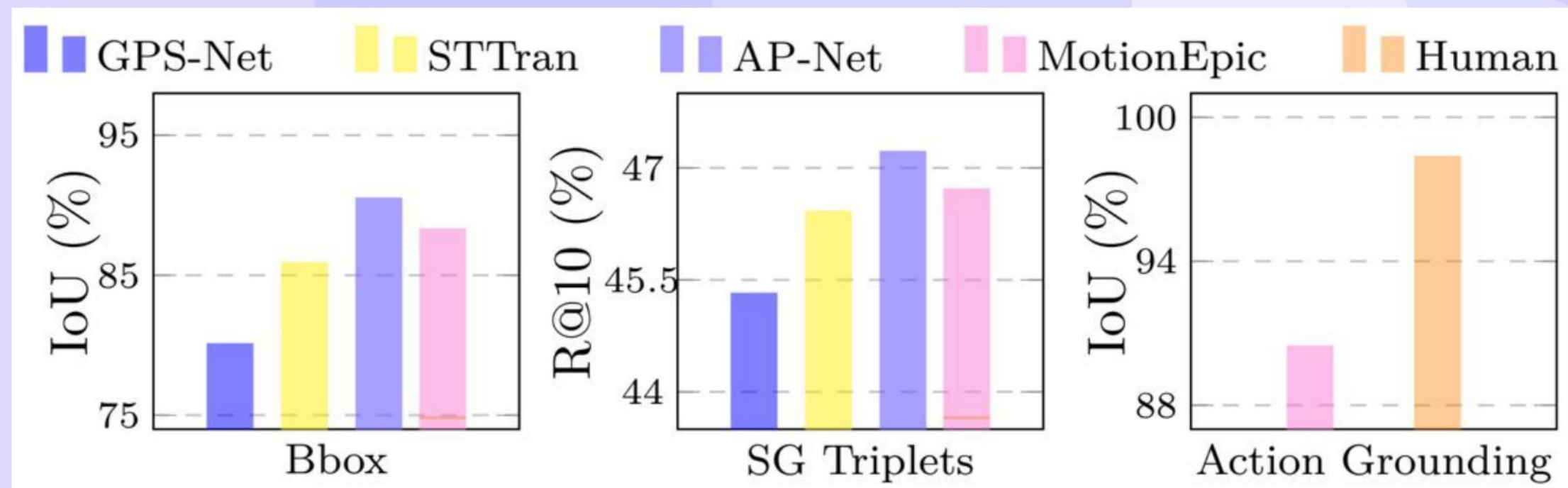
- Model can generate partial STSG as rationale
- Grounding-aware tuning enables STSG-free downstream inference
- Coarse-to-fine objectives align video and STSG for accurate parsing
- **Recycle**: generated structure feeds subsequent reasoning rounds
- **Outcome**: factual grounding reduces hallucination in answers



- **Overall:** VoT plus MotionEpic outperforms state of the art across complex VideoQA
- STSG integration and VoT surpass vanilla CoT approaches
- Implicit STSG integration in MotionEpic beats explicit feature fusion
- **Consistent gains:** fine-tuning and zero-shot settings
- CoT alone shows limited lift for video reasoning compared to VoT

- Large margins on causal and complex reasoning subsets
- Benefits in description, explanation, prediction, and counterfactual
- Consistent gains across Acc@All, causal, temporal, and descriptive splits
- **Stronger**: reasoning about interactions and intentions
- Stable improvements across multiple baselines and settings

- VoT beats CoT in zero-shot; verification boosts robustness
- Grounding verification critical for perceptive datasets; both matter for complex videos
- Dual verification yields clearest gains on complex QA (NExT-QA, STAR)
- **Finding:** VoT benefits grow when no in-domain training is used
- **Net effect:** higher accuracy through evidence-backed answers

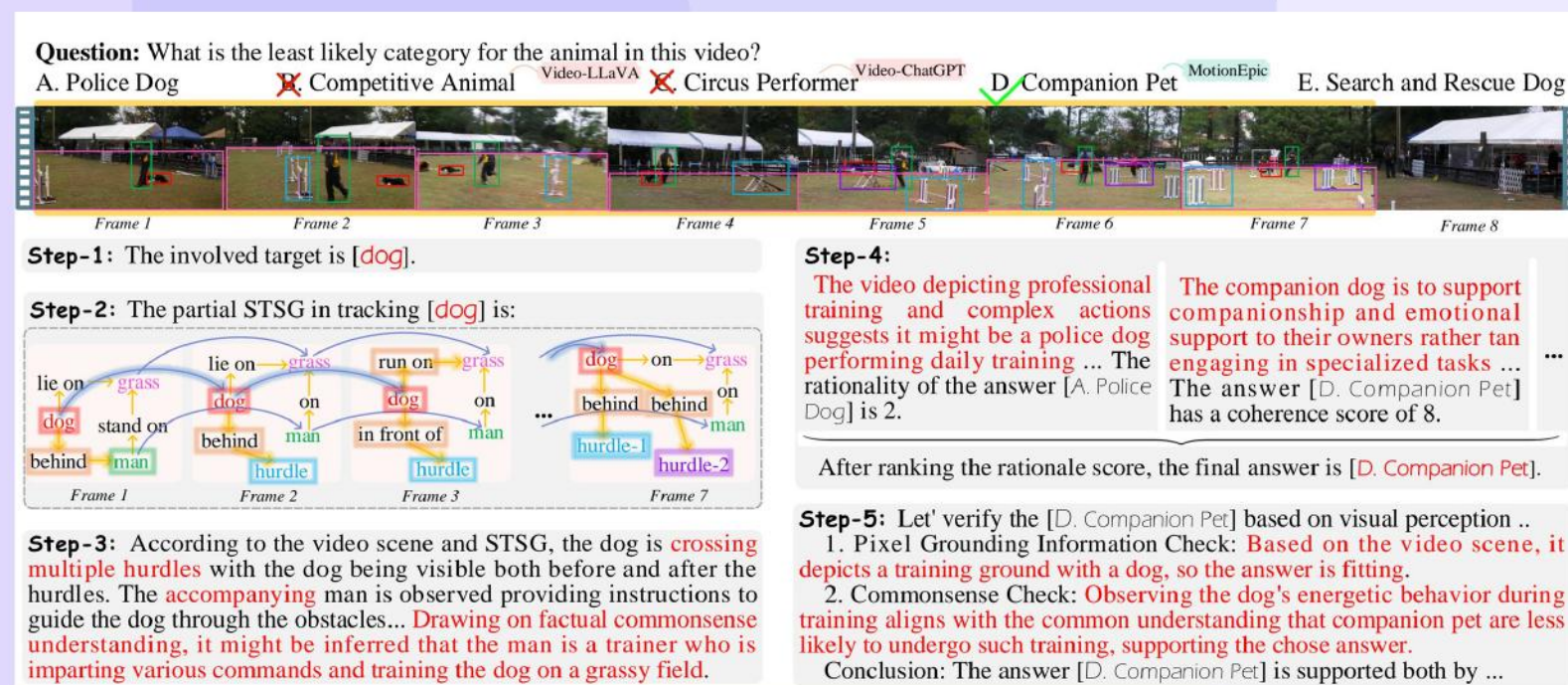
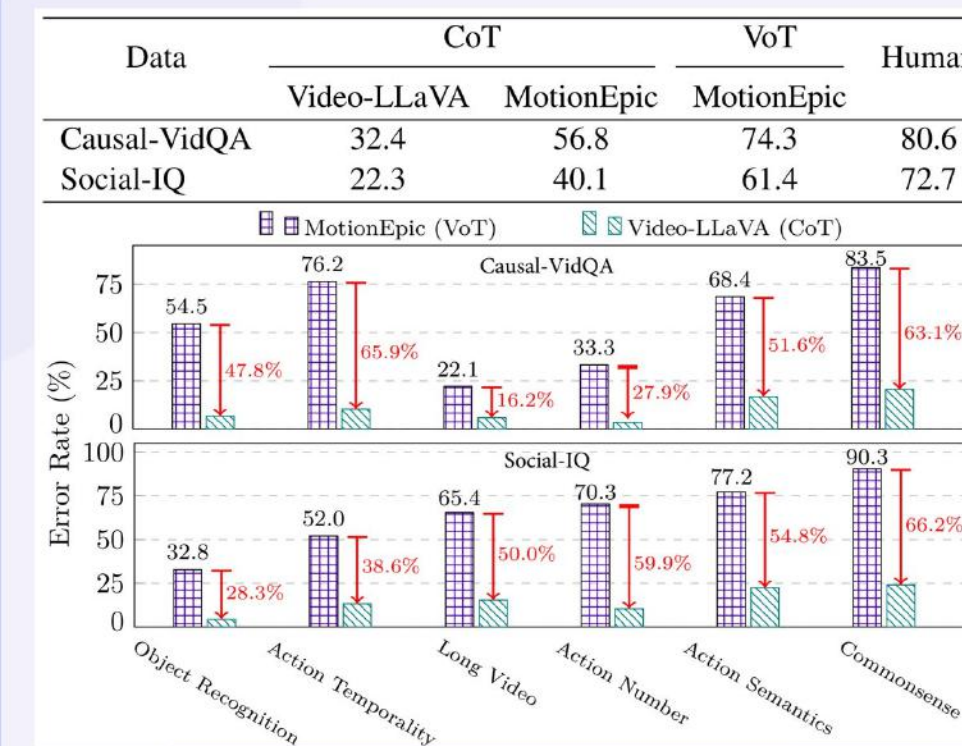


- MotionEpic competitive with STSG parsers on grounding tasks
- Ablations reveal which grounding objectives matter most
- Reliable grounding supports downstream multi-step reasoning

- **Evidence:** strong object, relation, and action alignment
- **Takeaway:** grounding quality correlates with reasoning accuracy

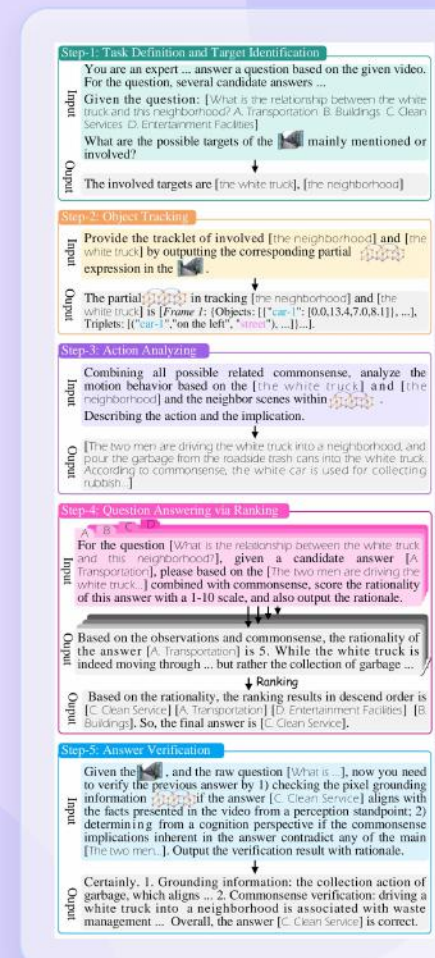
Lin et al., 2020; Cong et al., 2021; Li et al., 2022b

- MotionEpic plus VoT approaches human performance on hard subsets
- Error reductions in action semantics and commonsense
- Human study validates VoT gains over CoT baselines
- **Improved:** interpretability and answer faithfulness
- **Focus:** fewer mistakes in multi-entity interactions



- STSG grounding reduces hallucination
- Sequential reasoning and verification yields correct answer
- Explainable chain from perception to cognition

- **Rationale:** grounded entities, relations, and actions
- **Clarity:** traceable steps support human auditing



- **VoT:** first video-native CoT framework with dual verification
- **MotionEpic:** STSG-grounded video MLLM enabling STSG-free inference
- State-of-the-art across complex VideoQA; broad implications
- **Outlook:** transferable to richer multimodal reasoning tasks
- Bridging perception and cognition for reliable video understanding

Thank You

Questions & Discussion